# Dev Trusharkumar Shah

*M.S. Data Science candidate at UMD (4.0 GPA) with experience in NLP and distributed data systems — results measured against real baselines.*

shahdev5203@gmail.com · LinkedIn · GitHub · delvitron1019.github.io

## EDUCATION

**University of Maryland** — College Park, MD
*M.S. in Data Science | GPA: 4.0 / 4.0* — *Aug 2025 – May 2027*
Coursework: Machine Learning, Big Data Systems, Probability & Statistics, Algorithms for Data Science

**Indian Institute of Information Technology, Lucknow** — Lucknow, India
*B.Tech in Computer Science & AI | GPA: 8.78 / 10* — *Nov 2021 – May 2025*
Coursework: Data Structures & Algorithms, AI, NLP, Generative AI, DBMS, Computer Networks

## TECHNICAL SKILLS

**Languages:** Python, SQL, C++, R, Solidity

**ML & NLP:** PyTorch, TensorFlow, Hugging Face Transformers, BERT, T5, scikit-learn, XGBoost, YOLO

**Data & Infrastructure:** PySpark, Ray, Parquet, ETL pipelines, Docker, Git, AWS (S3, EC2), Flask REST APIs, Streamlit

**Statistics & Analytics:** A/B testing, hypothesis testing, bootstrap confidence intervals, experiment design, Power BI, Tableau

**Competitive Programming:** CodeChef 3-star (delvitron1019) — peak 1691, best contest global rank 39 | LeetCode top 5%, best contest rank 931

## EXPERIENCE

**Georgia Institute of Technology – Office of Development** — Atlanta, GA
*AI Strategy & Operations Intern* — *Jun 2026 – Jul 2026*
- Interviewed stakeholders across 12 Advancement teams to map manual workflows spanning 7 enterprise systems including Blackbaud CRM, ServiceNow, Workday, and Microsoft 365.
- Designed 14 AI automation solutions with end-to-end architectures, scoped to existing Microsoft tooling (Power Automate, Copilot Studio, Entra ID, Power BI) to avoid new software spend.
- Sequenced recommendations into a 3-phase adoption roadmap and presented a 22-slide assessment to leadership, covering data-governance constraints for enterprise Copilot deployment.

**Nativebyte** — New Delhi, India
*Software Developer Intern* — *Dec 2024 – May 2025*
- Built Python recommendation systems using XGBoost and regression models to improve UI personalization, and deployed end-to-end ML pipelines covering data cleaning, feature engineering, training, and cross-validation.

**IIIT Lucknow – AI Research Lab** — Lucknow, India
*AI Research Intern · Dr. Deepak Kumar Singh* — *Jan 2024 – May 2024*
- Benchmarked YOLOv3–v8 and Faster R-CNN in PyTorch and TensorFlow on Raspberry Pi, improving inference speed by 12%; profiled the accuracy/latency tradeoff to select configurations deployable on the target hardware, not merely highest-scoring offline.

**IIIT Allahabad – Research Lab** — Remote
*Software Engineering Research Intern · Dr. Naveen Saini* — *Aug 2023 – Nov 2023*
- Built a knowledge graph recommender via collaborative filtering on 10K+ multilingual ratings (18 languages); PCA and clustering improved precision by 9%, deployed via Flask REST API.

## KEY PROJECTS

**Phishing Email Detection** — *Python, BERT, Hidden Markov Models, scikit-learn*
- Diagnosed test-set leakage (HMMs and vocabulary fit before the split), rebuilt the pipeline leak-free, and re-measured on all 29,767 emails: 98.89% accuracy, 95% CI [98.62, 99.14], vs a 53.1% baseline.
- Ran the ablation the original omitted: 3 HMM-derived features matched 768 BERT dimensions with no separable difference (n=5,954), making the transformer removable from serving.

**Medical Text Summarization** — *Python, PyTorch, T5, Transformers, Streamlit*
- Fine-tuned T5 on 6,000 PubMed article/abstract pairs, beating a lead-3 extractive baseline by 1.84 ROUGE-L (95% CI [1.06, 2.57], n=400) and the untuned model by 5.91, with significance established by a paired bootstrap over 10,000 resamples; served via Streamlit.

**Distributed Log Processing** — *PySpark, pandas, Parquet*
- Benchmarked Spark against an identical single-threaded pandas implementation, locating the crossover between 1M and 4M rows (Spark 1.72x faster at 4M; pandas peak memory 608 MB → 1,959 MB), using a Zipf/log-normal generator to surface partition skew that uniform data hides.

**Patient-Controlled Medical Records** — *Solidity, Hardhat, OpenZeppelin*
- On-chain access control storing only pointers to encrypted off-chain payloads, keeping health data out of immutable storage; 25/25 tests passing, every operation gas-profiled.